

Lecture 1: Introduction

Michael Dinitz

September 1, 2026

601.433/633 Introduction to Algorithms

Welcome!

Introduction to (the theory of) algorithms

- ▶ How to design algorithms
- ▶ How to analyze algorithms

Prerequisites: Data Structures and MFCS/Discrete Math

- ▶ Small amount of review next lecture, but should be comfortable with asymptotic notation, basic data structures, basic combinatorics and graph theory.
- ▶ Undergrads from prereqs.
- ▶ “Informal” prerequisite: *mathematical maturity*

Instructors: Michael Dinitz, Jessica Sorrell

About me

- ▶ 11th time teaching this class (Fall 2014 – Fall 2026).
 - ▶ I'm still learning – let me know if you have suggestions!
 - ▶ Fall 2022: Sabbatical at Google Research-New York
 - ▶ Fall 2023: Parental leave
- ▶ Research in theoretical CS, particularly algorithms: approximation algorithms, graph algorithms, distributed algorithms, online algorithms.
- ▶ Also other parts of math (graph theory) and CS theory (algorithmic game theory, complexity theory) and theory of networking.
- ▶ Recently working a lot on *Differential Privacy* (601.438/638) and *Learning-Augmented Algorithms* (601.734)
- ▶ Office hours: TBD.

Administrative Stuff

- ▶ TA: Shruthi Prusty (CS PhD student). Office hours TBD
- ▶ CAs: Approximately six, still finalizing.
- ▶ Website: <https://introalgorithmsfall26.cs.jhu.edu>
 - ▶ Syllabus, schedule, lecture notes, office hours, ...
 - ▶ Courselore for discussion/announcements
 - ▶ Gradescope for homeworks/exams.
- ▶ Textbook: CLRS (fourth edition)

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- ▶ Textbook: CLRS (fourth edition)
- ▶ Class a bit different than in the past!
 - ▶ More but lower-value homeworks, more midterms.
 - ▶ Office hours (hopefully) M-Th, 6pm–9pm
 - ▶ Recitations (hopefully) Wednesdays

Grading and Exams

Grading: 20% homework, 20% midterm 1, 20% midterm 2, 40% final exam.

- ▶ “Curve”: Historically, average $\approx$ B+. About 50% A's, 50% B's, a few below.
 - ▶ Curve only helps! Someone else doing well does not hurt you.
 - ▶ Be collaborative and helpful (within guidelines).

Grading and Exams

Grading: 20% homework, 20% midterm 1, 20% midterm 2, 40% final exam.

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Exams: two midterms, final.

- ▶ Midterms: In-class (75 minutes), traditional, closed book
- ▶ Final: in person, Dec 15 (scheduled by registrar). 3 hours, traditional, closed book.

Homeworks

- ▶ Every week, posted on website (HW1 out next Tuesday).
 - ▶ Out on Tuesdays, due by 11:59pm the following *Monday*
- ▶ *Must* be typeset ($\text{\LaTeX}$ preferred, not required)
- ▶ Unlimited collaboration, but *individual* writeups.
 - ▶ Work together at a whiteboard to solve, then write up yourself.
- ▶ 120 late hours (5 late days) total
- ▶ Some homeworks (approximately 3-ish) will involve oral presentations instead of written solutions.
 - ▶ Experiment, still figuring out the details!
 - ▶ Groups of three
 - ▶ Probably starting with HW2

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- ▶ Cheating includes:
 - ▶ Collaborating with people outside your group of three.
 - ▶ Collaborating *with* your group on the writeup.
 - ▶ Looking online for the solutions/ideas to the problem *or related problems*, rather than to understand concepts from class.
 - ▶ Using ChatGPT or other LLMs.
 - ▶ Using Chegg, CourseHero, your friends, . . . , to find back tests, old homeworks, etc.
 - ▶ etc.

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- ▶ Just solve the problems with your group and write them up yourself!
 - ▶ Use the internet, classmates, other resources to understand concepts from class, not to help with assignments.
- ▶ In previous years, punishments have included zero on assignment, grade penalty, mark on transcript, etc. ≥ 1 person has had PhD acceptance revoked.

LLMs

LLMs can be very useful tools!

- ▶ Often know the right answer, particularly for “classical” settings like algorithms. Lots of training data!
- ▶ If you overuse them, you will not actually learn the material
- ▶ If you’re no better than an LLM, why should anyone hire you or care what you have to say?

Often incorrect, and will double down on mistakes

On Learning

- ▶ Learning is challenging
- ▶ The challenge is typically proportional to *how much you're learning*
- ▶ You are rewiring your brain to be able to think in new ways! In this class in particular, you are practicing convergent creativity, thinking precisely, rigorously, adversarially, recursively, etc. These are broadly useful skills!
- ▶ No amount of Professor Sorrell and me talking at you can give you these skills. You can think of the course staff like personal trainers. We can recommend exercises, demonstrate technique, but we can't do them for you. At the end of the day, if you want your brain to adapt to the challenges of this course, you need to practice (the more consistently, the better!)
- ▶ Practice should be challenging, but not painful! If you're struggling, please talk to us. We're here to help you succeed!

Course Overview

- ▶ Introduction to *Theory* of Algorithms: math not programming.
- ▶ Two goals: how to *design* algorithms, and how to *analyze* algorithms.
 - ▶ Sometimes focus more on one than the other, but both important

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 - ▶ Computational problem: given input $\mathbf{X}$, want to output $\mathbf{f}(\mathbf{X})$. How to do this?
- ▶ Things to prove about an algorithm:
 - ▶ Correctness: it does solve the problem.
 - ▶ Running time: worst-case, average-case, worst-case expected, amortized, ...
 - ▶ Space usage
 - ▶ and more!

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 - ▶ Space usage
 - ▶ and more!
- ▶ This class: mostly correctness and asymptotic running time, focus on worst-case

Why analyze? Why worst case?

- ▶ Obviously want to prove correctness!
 - ▶ Testing good, but want to be 100% sure that the algorithm does what you want it to do!

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 - ▶ Especially if your algorithm is “low-level”, it will be used in many different settings.

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Hard to determine experimentally.

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 - ▶ Especially if your algorithm is “low-level”, it will be used in many different settings.
- ▶ We will focus on how an algorithm “scales”: how running times change as input grows. Hard to determine experimentally.
- ▶ Most importantly: want to *understand*.
 - ▶ Experiments can (maybe) convince you that something is true. But can't tell you why!

Example 1: Multiplication

Multiplication I

Often an obvious way to solve a problem just from the definition. But might not be the right way!

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How to do this?

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Definition of multiplication:

- ▶ Add X to itself Y times: $X + X + \dots + X$. Or add Y to itself X times: $Y + Y + \dots + Y$.

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- ▶ $\Theta(Y)$ or $\Theta(X)$ (assuming constant-time adds).
- ▶ Could be $\Theta(2^n)$. *Exponential* in size of input ($2n$).

Multiplication II

Better idea?

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$$\begin{array}{r} 110110 = 54 \\ x 101001 = 41 \\ \hline 110110 \\ 110110 \\ + 110110 \\ \hline 100010100110 = 2 + 4 + 32 + 128 + 2048 = 2214 \end{array}$$

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Running time:

- ▶ $O(n)$ column additions, each takes $O(n)$ time $\implies O(n^2)$ time.
- ▶ Better than obvious algorithm!

Multiplication III

Can we do even better?

Multiplication III

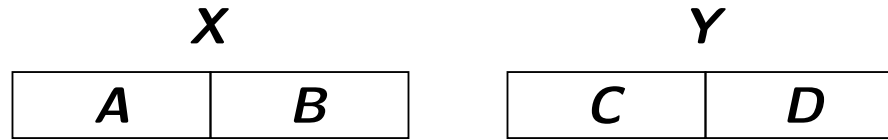
Can we do even better? Yes: *Karatsuba Multiplication*

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$$X = 2^{n/2}A + B$$

$$Y = 2^{n/2}C + D$$

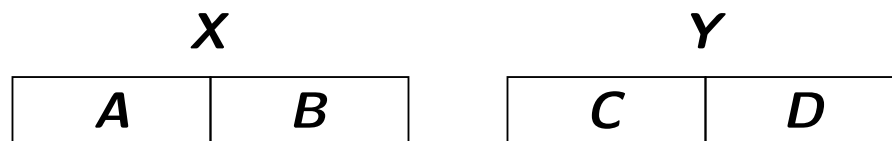


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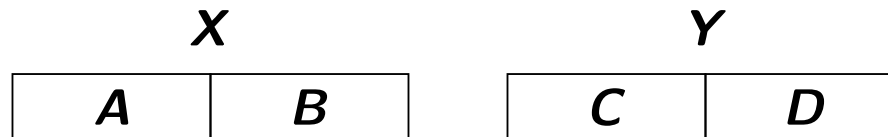
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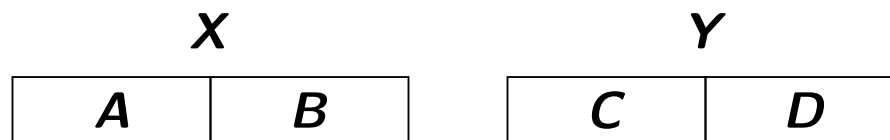
$$\begin{aligned}XY &= (2^{n/2}A + B)(2^{n/2}C + D) \\ &= 2^n AC + 2^{n/2}AD + 2^{n/2}BC + BD\end{aligned}$$

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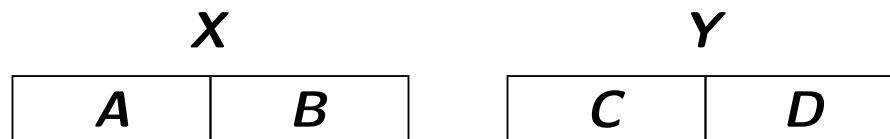
Four $n/2$ -bit multiplications, three shifts, three $O(n)$ -bit adds.

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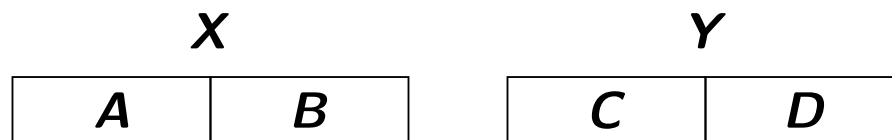
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Running Time: $T(n) = 4T(n/2) + cn \implies T(n) = O(n^2)$

Karatsuba Multiplication

Rewrite equation for XY :

$$\begin{aligned} XY &= 2^n AC + 2^{n/2} AD + 2^{n/2} BC + BD \\ &= 2^{n/2}(A + B)(C + D) + (2^n - 2^{n/2})AC + (1 - 2^{n/2})BD \end{aligned}$$

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Three $n/2$ -bit multiplications, $O(1)$ shifts and $O(n)$ -bit adds.

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Three $n/2$ -bit multiplications, $O(1)$ shifts and $O(n)$ -bit adds.

$$\implies T(n) = 3T(n/2) + c'n$$

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$$\implies T(n) = 3T(n/2) + c'n$$

$$\implies T(n) = O(n^{\log_2 3}) \approx O(n^{1.585})$$

Even Better Multiplication?

Can we do even better than Karatsuba?

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Theorem (Schönhage and Strassen)

There is an $O(n \log n \log \log n)$ -time algorithm for multiplication.

Uses Fast Fourier Transform (FFT).

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Theorem (Harvey and van der Hoeven '19)

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Example 2: Matrix Multiplication

Matrix Multiplication: Definition

Given $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n \times n}$, compute $\mathbf{XY} \in \mathbb{R}^{n \times n}$

- ▶ $(\mathbf{XY})_{ij} = \sum_{k=1}^n \mathbf{X}_{ik} \mathbf{Y}_{kj}$
- ▶ Don't worry for now about representing real numbers
- ▶ Assume arithmetic operations (multiplication, addition) in $\mathbf{O(1)}$ time

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Algorithm from definition:

- ▶ For each $i, j \in \{1, 2, \dots, n\}$, compute $(\mathbf{XY})_{ij}$ using formula.

Handwritten red annotations illustrating the formula for matrix multiplication. It shows a matrix $\mathbf{X}$ with a horizontal line and an arrow pointing to a specific row i . Next to it is a vertical line with an arrow pointing to a specific column j . These are followed by an equals sign and a bracketed space, with a small 'x' above it, representing the dot product of row i of $\mathbf{X}$ and column j of $\mathbf{Y}$.

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Running time:

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Algorithm from definition:

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Running time:

- ▶ $O(n^2)$ entries, each entry takes n multiplications and $n - 1$ additions $\implies O(n^3)$ time.

Strassen I

Break $\mathbf{X}$ and $\mathbf{Y}$ each into four $(n/2) \times (n/2)$ matrices:

$$\mathbf{X} = \begin{array}{|c|c|} \hline \mathbf{A} & \mathbf{B} \\ \hline \mathbf{C} & \mathbf{D} \\ \hline \end{array}$$

$$\mathbf{Y} = \begin{array}{|c|c|} \hline \mathbf{E} & \mathbf{F} \\ \hline \mathbf{G} & \mathbf{H} \\ \hline \end{array}$$

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So can rewrite $\mathbf{XY}$:

$$\mathbf{XY} = \begin{array}{|c|c|} \hline \mathbf{AE + BG} & \mathbf{AF + BH} \\ \hline \mathbf{CE + DG} & \mathbf{CF + DH} \\ \hline \end{array}$$

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Recursive algorithm: compute eight $(n/2) \times (n/2)$ matrix multiplies, four additions

Strassen II

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Running time: $T(n) = 8T(n/2) + cn^2 \implies T(n) = O(n^3)$.

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Running time: $T(n) = 8T(n/2) + cn^2 \implies T(n) = O(n^3)$.

Improve on this?

Strassen III

$$XY = \begin{array}{|c|c|} \hline AE + BG & AF + BH \\ \hline CE + DG & CF + DH \\ \hline \end{array}$$

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$$M_1 = (A + D)(E + H)$$

$$M_4 = D(G - E)$$

$$M_7 = (B - D)(G + H)$$

$$M_2 = (C + D)E$$

$$M_5 = (A + B)H$$

$$M_3 = A(F - H)$$

$$M_6 = (C - A)(E + F)$$

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$$XY = \begin{array}{|c|c|} \hline M_1 + M_4 - M_5 + M_7 & M_3 + M_5 \\ \hline M_2 + M_4 & M_1 - M_2 + M_3 + M_6 \\ \hline \end{array}$$

Strassen IV

$$M_1 = (A + D)(E + H)$$

$$M_2 = (C + D)E$$

$$M_3 = A(F - H)$$

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Only seven $(n/2) \times (n/2)$ matrix multiplies, $O(1)$ additions

Strassen IV

$$M_1 = (A + D)(E + H)$$

$$M_2 = (C + D)E$$

$$M_3 = A(F - H)$$

$$M_4 = D(G - E)$$

$$M_5 = (A + B)H$$

$$M_6 = (C - A)(E + F)$$

$$M_7 = (B - D)(G + H)$$

$$XY = \begin{array}{|c|c|} \hline M_1 + M_4 - M_5 + M_7 & M_3 + M_5 \\ \hline M_2 + M_4 & M_1 - M_2 + M_3 + M_6 \\ \hline \end{array}$$

Only seven $(n/2) \times (n/2)$ matrix multiplies, $O(1)$ additions

Running time: $T(n) = 7T(n/2) + c'n^2 \implies T(n) = O(n^{\log_2 7}) \approx O(n^{2.8074})$.

Further Progress

- ▶ Coppersmith and Winograd '90: $O(n^{2.375477})$
- ▶ Virginia Vassilevska Williams '13: $O(n^{2.3728642})$
- ▶ François Le Gall '14: $O(n^{2.3728639})$
- ▶ Josh Alman and Virginia Vassilevska Williams '21: $O(n^{2.3728596})$
- ▶ Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou '24: $O(n^{2.371552})$
- ▶ Josh Alman, Ran Duan, Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou '25: $O(n^{2.371339})$
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Is there an algorithm for matrix multiplication in $O(n^2)$ time?

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If you answer this (with proof!), automatic A+ in course and PhD

See you Thursday!